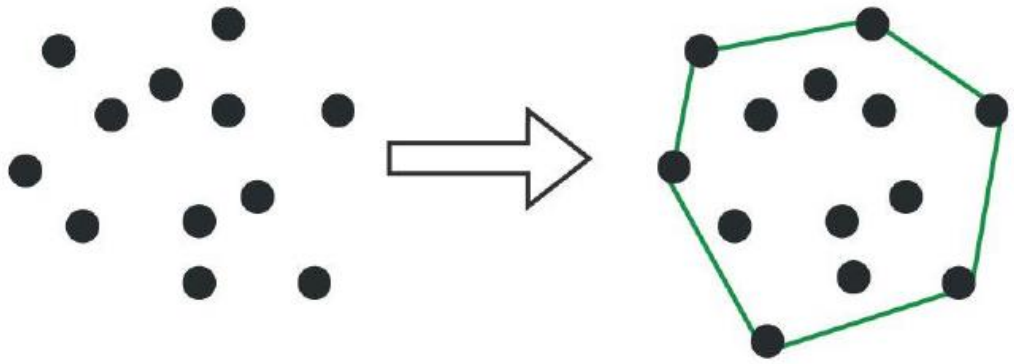
SARDAR VALLABHBHAI NATIONAL INSTITUTE OF TECHNOLOGY, SURAT
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
CS103: DESIGN AND ANALYSIS OF ALGORITHMS
M.Tech. (DS) Ird Year
Lab Assignment

Assignment 1:
Design a divide-and-conquer algorithm for computing the number of levels in a binary tree. (In particular, the algorithm must return 0 and 1 for the empty and single-node trees, respectively.) Analyze the time efficiency class of your algorithm. Now, implement the algorithm you designed and find out the time taken for empty, one-level, two-level and three-level binary trees. Compare it with the theoretical analysis that you would have done before.

Assignment 2:
Write programs to count the number of inversions in a given array using the brute force approach and using the algorithm applying the merge-sort discussed in the class (with time complexities O(n2) and O(n lg n) respectively). Compare both the programs with timing analysis with input size starting with n = 10 upto n = 105. Comment on the results obtained.

Assignment 3:
Observe that the algorithm PARTITION() for Quicksort given in the text is different from the one discussed in the class. Implement two different Quicksort routines that use the two different PARTITION routines. Test your implementation on an input vector of size at least 2048 integers and time the programs. Analyze the result and document it.

Assignment 4:
The convex hull of a set of points in a Euclidean space is the smallest convex polygon that encloses all of the points. In two dimensions (2D), the convex hull is a convex polygon. Given an array of coordinates x, y on a 2-D plain, your task is to print the set of coordinates that form the convex hull. Examples:



Input: points[] = {(0, 0), (0, 4), (-4, 0), (5, 0), (0, -6), (1, 0)}
Output: (-4, 0), (5, 0), (0, -6), (0, 4)
Input: points list = {{1, 2}, {3, 1}, {5, 6}}
Output: {{1, 2}, {3, 1}, {5, 6}}

Assignment 5:
Solve the shortest distance between two points in 3D. [hint: You can use a similar approach with a few modifications that you have used in the 2D version of this problem]

Assignment 6:
Do some research and find out what factors the actual time a program takes depends on. Use these factors as an argument to justify why it is not feasible to do empirical analysis when the goal is to compare two algorithms.

Assignment 7:
Generalize the 1D peak finding problem to 2D. A peak in a 2D matrix is an element greater than or equal to all its neighbours. Your task is to find such a peak in a matrix.
Constraints:

$$\mathbf{1 \leq n,m \leq 10^5}$$